

Chapter 43: Kernel and image Worksheet Worksheet

1. Are each of the following statements true or false?

- (a) If $T : V \rightarrow W$ is a linear transformation with trivial kernel (i.e. $\ker(T) = \{\mathbf{0}\}$), and $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a linearly independent set in V , then $\{T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_n)\}$ a linearly independent set in W .
- (b) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(\mathbf{x}) = \begin{bmatrix} 3 & 1 \\ 4 & 4 \end{bmatrix} \mathbf{x}$ has a trivial kernel (i.e. $\ker(T) = \{\mathbf{0}\}$).
- (c) If $T : V \rightarrow W$ is a linear transformation, then $\dim(\text{im}(T)) \leq \dim(V)$.

2. Consider a linear transformation $T : M_{22} \rightarrow \mathbb{R}^7$. What are the possible dimensions for $\ker(T)$ and $\text{im}(T)$? For extra practice, exhibit examples of linear transformations with all the possible dimensions.

3. Consider a linear transformation $T : M_{22} \rightarrow P_2$. What are the possible dimensions for $\ker(T)$ and $\text{im}(T)$? For extra practice, exhibit examples of linear transformations with all the possible dimensions.

4. Define $L : P_1 \rightarrow P_2$ by $L(p(x)) = xp(x) + p(0)$. Note: The previous worksheet asked you to show this is a linear transformation.

(a) Find $\ker(L)$.

(b) What is the dimension of the image of L ? Find a basis for the image of L .

5. Consider the function $L : M_{22} \rightarrow M_{22}$ by $L(A) = A + A^T$.

- (a) Show L is a linear transformation. (*Hint: You can use the properties $(A + B)^T = A^T + B^T$ and $(cA)^T = cA^T$*)
- (b) Find a basis for the kernel of L . What's the dimension?
- (c) Find a basis for the image/range of L . What's the dimension?

6. Consider $T : P_2 \rightarrow \mathbb{R}^3$ by $T(p) = \begin{bmatrix} p(1) \\ p'(1) \\ 0 \end{bmatrix}$.

- (a) Show that T is linear.
- (b) Find a basis for the kernel of T .
- (c) Find a basis for the image of T .